A Compute and Environment Configuration

All experiments were conducted on an NVIDIA A100 GPU with CUDA version 12.0, running on an Ubuntu 20.04.6 LTS server.

B Dataset Repository

We have provided a anonymous dataset repository for MedMKG, available at https://github.com/XiaochenWang-PSU/MedMKG/. The MedMKG dataset can be loaded alongside the MIMIC-CXR dataset, which requires separate download following the instructions provided in the repository README file. The repository also includes runnable code for data processing, baseline models, environment configuration, and example execution scripts. When using the data, users must adhere to the consent terms specified by the MIMIC-CXR maintainers.

C Deployment and Updating

MedMKG supports three categories of updates:

(1) **Foundational knowledge updates** (e.g., incorporating new UMLS releases).
(2) **Imaging dataset updates** (e.g., newly added MIMIC–CXR studies or revised radiology reports).
(3) **Multimodal extensions** (e.g., integration of CT, MRI, ultrasound, or EHR-derived features).

Because the construction pipeline is highly efficient, typically requiring only a few hours with API-based processing, the entire workflow can be re-executed whenever new data or modalities become available, thereby enabling continuous and real-time maintenance of MedMKG.

D Details of Knowledge Graph Construction

D.1 Pre-processing of MIMIC-CXR

To ensure the quality of the constructed multimodal knowledge graph, we perform targeted pre-processing on the raw data in the MIMIC-CXR database. Each radiological report may correspond to images in different views, including anteroposterior, posteroanterior, lateral, etc. Involving multiple images with the same set of concepts could result in significant redundant edges within the knowledge graph. Therefore, we only maintain images in the anteroposterior view for graph conciseness; similarly, radiological reports usually contain abundant information such as diagnostic history that is not directly relevant to the content of the corresponding radiological image, therefore, extracting concepts from these similar reports can also result in redundancy.

To mitigate this problem, we only preserve sections of Impression and Findings, two major sections that contain the most informative content, and stick to existing works in clinical report analysis [43]. We perform semantic filtering using DBSCAN [18] and MedCSPCLIP [71]. To be specific, we encode all the radiological reports with the text encoder of MedCSPCLIP, then perform clustering on the reports based on their semantics. Based on the clustering results, we select the ones near the centroid of each cluster as representative of a group of similar radiological reports.

These approaches function together, ensuring that our pipeline referred to in Section 3 receives high-quality data for processing, producing the multimodal knowledge graph with sufficient information, negligible noise, and minimal redundancy.

D.2 Filtering per Semantic Type of Medical Concepts

In order to eliminate concepts that are overly abstract or lack practical value, we filter concepts based on their semantic types. The following semantic types are not preferred and thus filtered, while all other semantic types in the UMLS vocabulary [10] are allowed. Here are the filtered types: *Occupation or Discipline, Intellectual Product, Age Group, Biomedical Occupation or Discipline, Classification, Patient or Disabled Group, Organization, Regulation or Law, Geographic Area, Health Care Related Organization, Language, Conceptual Entity, Professional Society, Group Attribute, Idea or Concept, Self-help or Relief Organization, Group, Temporal Concept, Professional or Occupational Group, Qualitative Concept, Quantitative Concept, Population Group, Functional Concept, Body System, and Family Group.*

D.3 Prompt for Concept Disambiguation and Relation Extraction

To leverage the LLM's contextual understanding for effective concept disambiguation and relation extraction, we designed an instructive prompt that guides the model through these tasks. The prompt is presented in Figure 3.

D.4 Selection of LLM

In this study, GPT-4o [52] is selected for disambiguation, as prior research has demonstrated its superior performance in biomedical comprehension [12, 59] and its effectiveness in resolving medical terminology ambiguity [32], compared with other LLMs. To further substantiate this choice, we conduct a case study to evaluate the suitability of GPT-4o for curating MedMKG, as shown in Figure 4.

The analysis reveals that GPT-4o outperforms other advanced LLM backbones, i.e., Gemini-2.5 [11] and LLaMA-3.1-8B-Instruct [23], by extracting more accurate concepts of interest while generating fewer hallucinations. These findings reinforce our decision to adopt GPT-4o for concept and relation extraction.

D.5 NaF Algorithm

We propose the Neighbor-Aware Filtering (NaF) algorithm for effective image filtering to boost the conciseness of MedMKG. More details are presented in Algorithm 1.

D.6 Illustration of MedMKG

Figure 5 shows the distribution of semantic types between the clinical concepts involved, indicating a broad and balanced coverage of the areas of clinical knowledge. Figure 6 shows a subgraph of MedMKG, provided to facilitate a better understanding of its structure and content. As shown in Figure 6, the medical multimodal knowledge graph integrates both intra- and cross-modal edges, offering rich multimodal medical knowledge that can potentially support a wide range of applications.

[10] https://www.nlm.nih.gov/research/umls/META3_current_semantic_types.html

Prompt for Concept Disambiguation and Relation Extraction

Report Text: [*Report Text*]

Candidate Concepts: [*Candidate Concepts*]

For each phrase, evaluate the concept candidates and select the most relevant concept based on the report context. Your decision should account for the radiological image context.

After selecting the concept for each phrase, classify its relation to the image:

- **Positive**: clearly represented in the image (e.g., anatomy, findings).
- **Neutral**: structural/general/meta/adjectives or not clinically informative (e.g., "findings", "normal", "changes").
- **Negative**: opposite of what is shown (e.g., concept implies pathology but image is normal).
- **Uncertain**: unclear presence (e.g., "possible", "could be").

Output constraint: Return only **Positive**, **Negative**, or **Uncertain**. Exclude **Neutral** entirely.

Output format:

```
***start***
(Concept ID only (starts with C), Relation)
***end***
```

Ensure that:

- Neutral concepts are excluded entirely.
- "findings" and other general/structural terms are treated as Neutral and omitted.
- Remaining concepts are checked against the report context and image relevance.

Figure 3: Prompt for concept disambiguation and relation extraction.

Entity & Relation Extraction Comparison Across LLMs

Radiological Report:
As compared to the previous radiograph, there is a further increase in extent of the opacities in the right lung. The left lung is constant. Changed nasogastric tube. Moderate cardiomegaly with extensive retrocardiac atelectasis.

Results:

Gemini-2.5

```
C0032285:Pneumonia, Uncertain
C0225706:Right Lung, Positive
C0029053:opacities, Positive
C0225730:Left Lung, Positive
C0018800:Cardiomegaly, Positive
C0004144:Atelectasis, Positive
```

LLaMA-3.1

```
C0032285:PNEUMONIA (Pneumonia), Positive
C0264716:Chronic heart failure, Positive
C0476273:Distress, Respiratory (Respiratory distress), Positive
```

GPT-4o

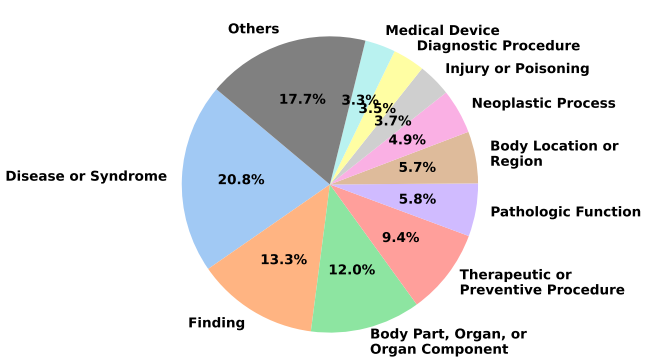
```
C0029053:opacities (Decreased translucency), Positive
C0225706:Right Lung (Right lung), Positive
C0085678:Nasogastric Tube (Nasogastric tube), Positive
C0018800:CARDIOMEGALY (Cardiomegaly), Positive
C0004144:ATELECTASIS (Atelectasis), Positive
```

Figure 4: Entity & relation extraction comparison across LLMs.

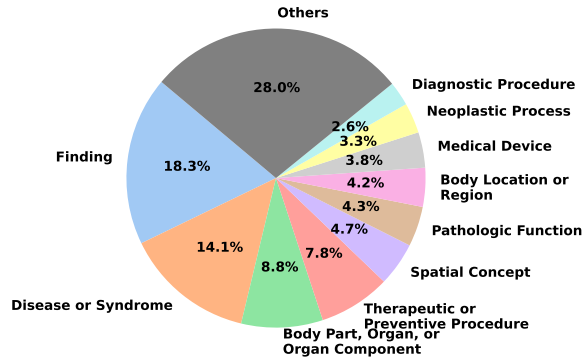
E Details of Human Assessment

E.1 Assessment Criteria

We conducted a human evaluation to assess the quality of MedMKG. Three key metrics were used:



(a) Distribution of Head Concepts per Semantic Types



(b) Distribution of Tail Concepts per Semantic Types

Figure 5: Distribution of entities involved in MEDMKG. The top 10 semantic types are shown individually, and rare types are grouped as "Others."

Algorithm 1 Neighbor-Aware Filtering Algorithm

Input:
- A set of images $\mathcal{M} = \{m_1, m_2, \ldots, m_N\}$.
- For each image m_i, its associated triplets $T_i = \{(m_i, r_{ij}, c_{ij})\}$.
- The set of filtered clinical concepts C.

Output: Selected image set $\mathcal{M}^*$.
$\mathcal{M}^* \leftarrow \emptyset$ and $C^* \leftarrow \emptyset$.
for each image $m_i \in \mathcal{M}$ **do**
  Compute $\text{Score}(m_i) \leftarrow \sum_{(r,c)\in T_i} \log \frac{M}{|N_{(r,c)}|}$.
end for
Sort $\mathcal{M}$ in descending order by $\text{Score}(m_i)$.
for each image m_i in sorted order **do**
  if $C^* \neq C$ **then**
    $\mathcal{M}^* \leftarrow \mathcal{M}^* \cup \{m_i\}$.
    $C^* \leftarrow C^* \cup \{c \mid \exists\, r \text{ such that } (r, c) \in T_i\}$.
  else
    break
  end if
end for
return $\mathcal{M}^*$.

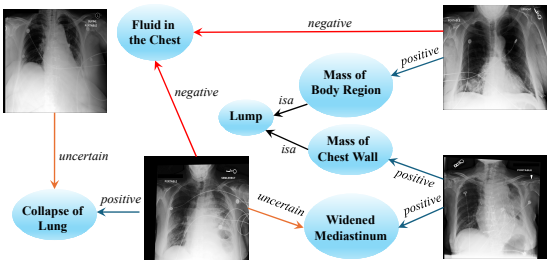


Figure 6: An illustration of MEDMKG.

- **Concept Coverage** measures how comprehensively the extracted concepts capture the clinically meaningful findings present in the image.
- **Relation Correctness** assesses whether the relationships between images and extracted concepts are accurately modeled, correctly identified with positive, negative, or uncertain associations.
- **Image Diversity** evaluates whether the set of images associated with each concept reflects a diverse range of clinical scenarios, rather than highly homogeneous ones.

These metrics were selected to capture complementary aspects of performance: *Concept Coverage* ensures clinical relevance and completeness; *Relation Correctness* ensures accurate representation of image-concept associations; and *Image Diversity:* ensures the robustness and generalizability of concept representations. Together, they provide a holistic evaluation of both precision and breadth of MEDMKG.

E.2 Assessment Procedure

For the metrics of concept coverage and relation correctness, we randomly sample 30 images in MEDMKG, choose all their concept neighbors, and the relation connecting them for assessment. For image diversity, we randomly choose 30 concepts in MEDMKG and provide all the images positively linked with them to the evaluator. The evaluator performs the assessment along with detailed guidance.

The guidebook is available in the repository.

F Details of Link Prediction

F.1 Implementation

All models are optimized using the AdamW optimizer [41] with a batch size of 2,048 and a learning rate of 0.001. The training is run for a maximum of 500 epochs with an early stopping mechanism (patience set to 5 epochs) to prevent overfitting. Data are split into training, validation, and test sets with an 8:1:1 ratio. We run each experiment with 5 random seeds and report the mean in Table 2.

F.2 Link Prediction Baselines

We benchmark MEDMKG with the following baseline models in the task of link prediction:

- **AttH** [8] is a hyperbolic knowledge graph embedding model designed to capture hierarchical structures by leveraging the Lorentz model.
- **DistMult** [80] is a bilinear factorization model for knowledge graphs that represents relations as diagonal matrices, enabling efficient computation.
- **TransR** [36] extends TransE by introducing separate relation-specific entity spaces, allowing better modeling of diverse relationships.
- **HypER** [3] applies hypernetworks to generate relation-dependent transformation matrices for entity embeddings, improving flexibility.
- **SimplE** [29] is an extension of Canonical Polyadic (CP) decomposition that enables each entity representation to be used in two different ways.
- **TuckER** [4] is based on Tucker decomposition and factorizes the knowledge graph tensor into entity and relation embeddings with a core interaction tensor.
- **MurP** [2] embeds knowledge graphs in the Poincaré ball model, enabling effective representation of hierarchical data.
- **MurE** [2] embeds knowledge graphs in Euclidean space using multiple relational constraints to improve predictive performance.
- **NTN** [60] introduces a neural tensor network for knowledge graph embedding, modeling entity interactions through a bilinear tensor layer.
- **TransD** [26] extends TransE and TransH by introducing entity- and relation-specific projection matrices for dynamic embedding transformation.
- **TransE** [7] models relationships as translations in the embedding space, assuming that the sum of the head and relation embeddings approximates the tail embedding.
- **RESCAL** [50] models multi-relational data using a bilinear tensor factorization approach that captures pairwise interactions.
- **RotatE** [61] represents relations as rotations in a complex vector space, capturing symmetric and antisymmetric relations effectively.
- **TransH** [75] introduces relation-specific hyperplanes to improve the representation of diverse relational properties.
- **ConvE** [15] applies 2D convolutional neural networks to entity embeddings, capturing complex interactions between entities and relations.
- **ComplEx** [63] extends DistMult by using complex-valued embeddings, enabling the representation of asymmetric relations.
- **ConvR** [27] applies relation-specific convolutional filters to entity embeddings, enhancing the modeling of complex interactions.

F.3 Evaluation Metrics

For the link prediction tasks, we use Mean Rank (MR) and Hits@K to evaluate the baselines. Lower MR and higher Hits@K indicate better performance.

G Backbone Models in Knowledge-augmented Tasks

The following advanced visual language models are adapted as the standard backbone for knowledge-augmented methods:

- **CLIP** [55] is a vision-language model trained on large-scale internet data using contrastive learning. It aligns images and text embeddings in a shared latent space, enabling zero-shot image classification and retrieval. The model is under the MIT License.
- **PubmedCLIP** [17] is a domain-specific adaptation of CLIP trained on PubMed articles and biomedical images. It enhances the alignment of biomedical images with textual descriptions, improving zero-shot performance in medical imaging tasks. The model is under the MIT License.
- **BioMedCLIP** [85] is a biomedical contrastive pretraining model trained on a large-scale corpus of biomedical images and text. It is designed to improve multimodal understanding in healthcare applications, particularly for retrieval and classification tasks. The model is under the MIT License.
- **MedCSPCLIP** [71] is a medical-specific adaptation of CLIP that incorporates the MedCSP framework for contrastive scalable pre-training. It learns generalizable medical image representations, enabling improved zero-shot performance and transfer learning in clinical applications. The model is under the MIT License.

H Details of Knowledge-augmented Image-text Retrieval

H.1 Datasets & Backbone Models

We leverage two representative datasets for the medical text-image retrieval task, i.e., OpenI [14] and MIMIC-CXR [28], following previous work [71]. To prevent any potential data leakage regarding MIMIC-CXR, we only select text-image pairs that were not used during the curation of MedMKG, and we randomly sample a fixed set of 10,000 pairs from these remaining examples. Since no predefined splits exist, both datasets are divided into training, validation, and test sets with an 8:1:1 ratio. To comprehensively assess the impact of knowledge augmentation, we employ four open-sourced vision–language models as backbones: CLIP [55], PubMedCLIP [17], BioMedCLIP [85], and MedCSPCLIP [71], as detailed in Appendix G.

H.2 Implementation

We comprehensively evaluate retrieval performance using standard metrics, i.e., precision@K and recall@K, with K set to 10, 20, and 100. Detailed metric descriptions can be found in Appendix H.4. All models are optimized using the AdamW optimizer [40]. The hidden state dimension is uniformly set to 512, and the learning rate is configured to 0.0001. Training is conducted for a maximum of 30 epochs with an early-stopping patience of 3 epochs.

H.3 Baselines

In the task of knowledge-augmented image-text retrieval, we benchmark with the following baseline models:

- **KnowledgeCLIP** [53]: This model extends CLIP by integrating external knowledge graphs. By adding knowledge-based objectives during pre-training, it leverages structured relational data to improve semantic alignment between images and text.

Table 6: Results (%) on knowledge-augmented visual question answering with PubMedCLIP as the backbone.

Methods	VQA-RAD				SLAKE				PathVQA			
	Acc↑	Prec↑	Rec↑	F1↑	Acc↑	Prec↑	Rec↑	F1↑	Acc↑	Prec↑	Rec↑	F1↑
MedCSPCLIP	68.13	61.59	**85.59**	71.63	66.20	83.95	38.86	53.12	77.72	73.37	**92.24**	81.73
+ KRISP	**80.08**	**84.00**	71.19	**77.06**	70.70	**91.76**	44.57	60.00	83.19	**94.71**	72.96	82.43
+ MKBN	69.72	65.44	75.42	70.08	67.32	75.21	50.29	60.27	85.37	86.17	86.84	86.51
+ K-PathVQA	67.73	75.34	46.61	57.59	71.55	74.03	65.14	69.30	85.31	89.35	82.65	85.87
+ EKGRL	76.10	73.39	77.12	75.21	69.30	78.95	51.43	62.28	84.92	92.75	78.19	84.85
+ MR-MKG	78.49	77.59	76.27	76.92	**83.94**	83.15	**84.57**	**83.85**	**86.53**	89.74	84.75	**87.17**
PubMedCLIP	66.14	64.35	62.71	63.52	63.94	59.59	**83.43**	69.52	81.26	86.65	77.20	81.65
+ KRISP	76.10	76.85	70.34	73.45	75.77	**79.47**	68.57	73.62	84.41	88.13	82.21	85.07
+ MKBN	67.33	67.31	59.32	63.06	70.70	75.18	60.57	67.09	84.56	90.15	80.18	84.87
+ K-PathVQA	72.51	76.34	60.17	67.30	68.17	67.03	69.71	68.35	83.76	87.62	81.44	84.42
+ EKGRL	76.49	75.21	74.58	74.89	75.49	73.40	78.86	76.03	**84.59**	**90.41**	79.96	84.86
+ MR-MKG	**78.88**	**76.86**	**78.81**	**77.82**	**77.75**	78.57	75.43	**76.97**	84.18	86.07	**84.36**	**85.21**

- **FashionKLIP** [73]: Designed for the fashion domain, FashionKLIP automatically constructs a multimodal conceptual knowledge graph from large-scale fashion data. By injecting domain-specific knowledge into the pre-training process, it learns fine-grained representations that enhance image-text alignment and retrieval performance.

H.4 Evaluation Metrics

For this task, we use Precision@k and Recall@k as evaluation metrics. A higher Precision@k indicates that a larger proportion of the retrieved items are relevant, whereas a higher Recall@k suggests that a greater proportion of all relevant items have been retrieved. These metrics together provide a comprehensive evaluation of the retrieval performance.

I Details of Knowledge-augmented Visual Question Answering

I.1 Datasets & Backbone Models

We compare the baselines on three medical visual question answering dataset, including VQA-RAD, SLAKE and PathVQA. We extract closed questions in these datasets for benchmarking. The same set of backbone models listed in Appendix G are applied on the datasets.

I.2 Implementations

We use the same implementation configuration as described in Appendix H.2.

I.3 Baselines

In the task of knowledge-augmented visual question answering, we evaluate five models that incorporate external knowledge graphs to improve visual reasoning and answer prediction:

- **KRISP** [45]: This model integrates structured knowledge graphs into the VQA pipeline, refining both image representations and question understanding to boost answer accuracy.
- **MKBN** [25]: Originally designed for medical VQA, MKBN leverages domain-specific knowledge graphs to align visual and textual features, thus enhancing performance in specialized settings.
- **K-PathVQA** [49]: By incorporating multi-hop reasoning over a knowledge graph, K-PathVQA enables the model to infer complex relationships and answer questions that require multi-step deductions.
- **EKGRL** [56]: This framework combines graph-based representation learning with reinforcement learning to effectively integrate external knowledge, thereby improving reasoning capabilities in visual question answering.
- **MR-MKG** [33]: MR-MKG utilizes contrastive loss to capture diverse semantic interactions between visual content and questions, leading to enhanced cross-modal alignment and VQA performance.

I.4 Evaluation Metrics

For the visual question answering task, we adopt four standard metrics: Accuracy, Precision, Recall, and F1 score.

I.5 Additional Experiment Results

In Table 6, we provide additional experiment results on the knowledge-augmented visual question answering task with the backbones of MedCSPCLIP and PubMedCLIP, which are not covered in Table 4 due to space limitations.

Consistent with Table 4, despite minor instability, incorporating MedMKG yields notable performance improvements across the baselines. This trend reinforces our conclusion that the graph's clinical specificity, image-aware relational structure, and semantic richness jointly contribute to stronger multimodal understanding.